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Research Paper

Accounting for repeat intervention costs in the economic comparison of laser interstitial thermal therapy and anterior temporal lobectomy for treatment of refractory temporal lobe epilepsy

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ABSTRACT

Objective: Laser interstitial thermal therapy (LITT) is an alternative to anterior temporal lobectomy (ATL) for the treatment of temporal lobe epilepsy that has been found by some to have a lower procedure cost but is generally regarded as less effective and sometimes results in a subsequent procedure. The goal of this study is to incorporate subsequent procedures into the cost and outcome comparison between ATL and LITT.

Methods: This single-center, retrospective cohort study includes 85 patients undergoing ATL or LITT for temporal lobe epilepsy during the period September 2015 to December 2022. Of the 40 patients undergoing LITT, 35 % (N = 14) underwent a subsequent ATL. An economic cost model is derived, and difference in means tests are used to compare the costs, outcomes, and other hospitalization measures.

Results: Our model predicts that whenever the percentage of LITT patients undergoing subsequent ATL (35% in our sample) exceeds the percentage by which the LITT procedure alone is less costly than ATL (7.2% using total patient charges), LITT will have higher average patient cost than ATL, and this is indeed the case in our sample. After accounting for subsequent surgeries, the average patient charge in the LITT sample ($103,700) was significantly higher than for the ATL sample ($88,548). A second statistical comparison derived from our model adjusts for the difference in effectiveness by calculating the cost per seizure-free patient outcome, which is $108,226 for ATL, $304,052 for LITT only, and $196,484 for LITT after accounting for the subsequent ATL surgeries.

Significance: After accounting for the costs of subsequent procedures, we found in our cohort that LITT is not only less effective but also results in higher average costs per patient than ATL as a first course of treatment. While cost and effectiveness rates will vary across centers, we also provide a model for calculating cost effectiveness based on individual center data.

1. Introduction

The most effective treatment for patients with medication-refractory mesial temporal lobe epilepsy (mTLE) is resection of the epileptogenic zone, with anterior temporal lobectomy (ATL) considered the gold standard. [1] Magnetic resonance imaging (MRI)-guided laser interstitial thermal therapy (LITT) is a minimally invasive alternative to ATL [2] that is growing in popularity [3,4] for several reasons, among them being a lower procedure cost, reduced length of stay (LOS), and fewer associated complications. [3,5,6] However, LITT is broadly only about 84.3 % as effective as ATL in achieving seizure freedom, [7] and the relative rates of effectiveness vary across surgical centers and are often substantially lower. [3,8].

Due to the reduced likelihood of achieving seizure freedom, a portion of LITT patients undergo a subsequent procedure, most commonly ATL. [3,7,9,10] Currently, none of the studies that compare the costs, outcomes, LOS, or other hospitalization data for LITT relative to ATL account for the combined costs and values for those LITT patients who

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undergo a subsequent ATL procedure, though authors have called attention to this issue as a necessary topic for future research. [3].

The failure to account for these subsequent procedures is problematic because if enough LITT patients have a second ATL procedure then any economic advantage from an initial lower procedure cost could be more than offset from the higher cost of having multiple procedures. This could mean that having an initial ATL might be both less costly and more effective on average than LITT. Similarly, other theorized advantages of LITT such as a reduced LOS for the average patient could also be more than offset accounting for the hospitalization time for patients having both procedures.

To date, there has not been an analysis that accounts for these subsequent procedures using actual patient data. The goal of this investigation is to develop an economic cost model that helps to predict when LITT will, or will not, have a cost advantage over ATL, and to statistically assess the relative advantages within a large sample of actual patients. We also derive a decision-criteria that can be applied to differing costs and effectiveness rates across surgical centers to predict which of the procedures is more cost effective in achieving seizure free outcomes once accounting for the likelihood of subsequent ATL procedures among LITT patients.

2. Methods

2.1. Economic cost and outcome models

To understand the relative cost of ATL versus LITT, including the cost of ATL revision surgery among some unsuccessful LITT patients, we derive an economic cost-benefit model. We first use the model to calculate the percentage of LITT patients undergoing subsequent ATL that results in the average per patient cost of LITT to be higher than the average cost for ATL alone. Second, we use the model to develop a decision-making guideline that can be used to determine whether LITT or ATL is expected to be more cost-effective per seizure-free patient outcome that can then be applied to any surgical center's relative costs and effectiveness rates of the two procedures. We subsequently apply these models to our sample of actual patient data.

Proposition 1. *The average cost per patient among LITT patients (including subsequent ATL procedures) will be* ***higher*** *than the average cost among ATL (only) patients if the proportion of LITT patients undergoing ATL exceeds the percentage by which LITT procedure costs are lower than ATL procedure costs. Mathematically this can be stated as*:

$$\frac{N_{L+A}}{N_L} > \frac{C_A - C_L}{C_A}$$

Where N_L is the number of patients undergoing LITT, N_{L+A} is the number of LITT patients undergoing a subsequent ATL, C_A is the average cost of an ATL procedure, and C_L is the average cost of a LITT procedure. We present the proof of Proposition 1 in Appendix 1A.

The first term, N_{L+A} / N_L is thus the percentage of total LITT patients undergoing subsequent ATL. Note this is the percentage of all LITT patients, not just the percentage of LITT patients for which LITT was ineffective (although it could be derived by dividing this percentage of total LITT patients by the failure rate of LITT to achieve seizure freedom).

The second term, (C_A – C_L) / C_A is the percentage by which the average cost of LITT is less than the average cost of ATL. This proposition can be restated inversely that the average cost per patient will be *lower* for LITT when the inequality sign in the above equation is reversed.

Thus, if 35 % of LITT patients undergo subsequent ATL, per patient costs will be lower among the group of LITT patients only if the initial LITT procedure is more than 35 % less costly than an ATL procedure. If, for example, 35 % of LITT patients undergo subsequent ATL but the cost of a LITT procedure is only 20 % less costly than ATL, the proposition implies that the LITT group will have a higher per patient costs than if all patients had gone only through ATL. We will examine this condition using our sample of patient data as well as other estimates from the literature.

The above condition enables the prediction of whether the average cost per patient will be higher or lower for LITT as a first surgical choice of treatment relative to ATL once accounting for the higher costs among LITT patients who undergo subsequent ATL surgery. However, this comparison does not account for the difference in the number of patients who become seizure free. A second, and perhaps more meaningful, statistic for comparison between the two procedures which we now derive is the expected cost *per seizure-free patient outcome* achieved.

This second model proposition rests upon the general economic principle that one wishes to minimize costs for a given level of benefit, [11] and this is the standard approach applied within the field of health economics to assess the cost effectiveness of all surgical interventions. [12].

Proposition 2. *The average cost per seizure-free patient outcome among LITT patients (including subsequent ATL procedures) will be* ***higher*** *than the average cost per seizure-free patient outcome among ATL (only) patients, once accounting for subsequent ATLs after LITT if*:

$$\frac{C_A}{SF_A} < \frac{C_L}{SF_L}$$

Where C_A and C_L are again the average cost of an ATL or LITT procedure respectively, SF_A is the seizure-free rate associated with ATL, and SF_L is the seizure-free rate associated with an initial LITT procedure (alone). We present the proof of this proposition in Appendix 1B. This proposition can be restated inversely that the average cost per seizure-free patient outcome will be *lower* for LITT when the inequality sign in the above equation is reversed.

The fractions or ratios on each side of the equation in Proposition 2 above have an easy verbal interpretation. They are the dollar costs of achieving one seizure-free patient outcome from each procedure. For example, if the seizure free rate among LITT patients is 50 %, and the average cost of a LITT procedure is $80,000, then, on average, it costs $160,000 to achieve one seizure-free patient using LITT ($80,000 /.50). This is because with a 50 % success rate, on average it takes two LITT surgeries to achieve one seizure-free patient outcome. If, accordingly, the seizure free rate among ATL patients is 80 %, and the average cost of an ATL procedure is $90,000 then, on average, it costs $112,500 to achieve one seizure-free patient using ATL ($90,000 /.80). Given the numbers in this example, ATL would be the more economically efficient means of achieving seizure freedom among a group of patients. We will examine this condition using our sample of patient data as well as other estimates from the literature.

A corollary of Proposition 2 can be obtained by reorganizing the terms in the equation. This version states that LITT will have a ***higher*** cost per seizure-free patient outcome (including the follow up ATL surgeries) when:

$$\frac{SF_L}{SF_A} < \frac{C_L}{C_A}$$

This corollary implies that LITT will be more costly per seizure-free patient outcome when the cost of a LITT procedure relative to an ATL procedure (the ratio of the average costs given by C_L / C_A) is greater than the relative effectiveness of LITT at achieving seizure freedom (proportionate to ATL, given by SF_L / SF_A). This also can be stated in its inverse that LITT will be less costly per seizure-free patient outcome when the relative effectiveness of achieving seizure freedom from LITT is greater than the relative cost of a LITT procedure (both relative to ATL).

As a numerical example, if the cost of an average LITT procedure is 80 % of the cost of the average ATL procedure, then LITT will be lower cost per seizure free outcome only if the seizure-free rate associated with LITT is more than 80 % of the seizure free rate associated with ATL. If instead the seizure-free rate associated with LITT is only half (50 %) as good as the seizure free rate associated with ATL, then at 80 % of the cost, LITT (with the associated follow up ATLs) is more costly in producing each seizure free outcome. We will also examine this condition using our sample of patient data as well as other estimates from the literature.

2.2. Data source

Our single-center, retrospective study examined patients undergoing either ATL or LITT for temporal lobe epilepsy during the period September 2015 (following inception of the use of LITT at our institution) to December 2022. The data was approved by the Institutional Review Board and did not require patient consent. Patient demographic, radiologic, and neurophysiologic data were recorded from retrospective chart review. Similarly, seizure outcomes were obtained by review of the most recent documented clinical encounter and classified according to Engel recommendations. [13] Patients were considered seizure-free (or good outcome) if Engel class I and were not seizure-free (poor outcome) if Engel class II-IV. Financial data were obtained via hospital accounting records, although these were unavailable prior to July 2016.

2.3. Inclusion and exclusion criteria

Patients were included if they were adults (18 or older), had a diagnosis refractory medial TLE, and underwent either a LITT or ATL procedure during the sample period. In total, we identified 109 patients who underwent a total of 123 surgeries. Three patients were excluded for the presence of medial temporal lobe tumor. Twenty-one patients were excluded due to lack of financial data. This resulted in 85 patients who underwent a total of 99 surgeries (59 ATL, 40 LITT). Among the 40 patients undergoing LITT, 35 % ($n = 14$) underwent a subsequent, though none of the patients who had an initial ATL underwent a subsequent surgery.

All patients are included in the cost comparison data. For analyzing the rate of seizure freedom, patients were excluded if they did not have a follow-up period of more than 6 months (unless they had a subsequent procedure prior to 6 months, which would imply failure, and were therefore included). Six patients undergoing seven surgeries were excluded from the seizure-freedom analysis, resulting in 79 patients undergoing 92 surgeries, 13 of whom had LITT followed by ATL.

2.4. Statistical analysis

Statistical analysis was completed using Stata MP version 14.1 (Stata Corp LP) and the Real Statistics Resource Pack software release 8.8.1. [14] The normality of all data distributions was tested using a Shapiro-Wilk test ($p \leq .05$). Due to non-normality, difference in means tests are performed using the Mann–Whitney *U* test (also called the Mann–Whitney–Wilcoxon test or Wilcoxon rank-sum test) which is a nonparametric difference in means test robust to outliers and non-normal distributions. Difference in means tests are performed assuming unequal variances and two independent samples, and all tests are two tailed with statistical significance defined as $p \leq .05$. For seizure-free outcome proportions, a two-sample two-tailed difference in proportions Z-test was performed to compare the outcomes among groups.

2.5. Surgical technique

2.5.1. Anterior temporal lobectomy

The patient is positioned supine with head rotated contralaterally. Next the head is secured in a Sugita head frame and a curvilinear incision is marked 1 cm anterior to the tragus to the level of the superior temporal line. The scalp flap is reflected with the fat pad and the temporalis is cut obliquely, in-line with its muscle fibers, which we retract with the Sugita skin hooks. A single burr hole is placed above the root of the zygoma, and a craniotomy is made in the standard fashion over the temporal lobe. Once the dura is opened, we measure the distance to the temporal tip and create a cortectomy in the superior temporal gyrus (roughly 4 cm dominant, 5.5 cm non-dominant). Subpial dissection inferior to the Sylvian fissure is carried out towards the temporal tip. The cortectomy is then carried inferiorly from our initial starting point down to middle fossa floor. Dissection is maintained from inferior to superior through the temporal white matter working medially until the temporal horn is encountered. The remaining temporal neocortex is removed *en bloc* for pathologic examination. The uncus is then removed in a subpial fashion working posteriorly until reaching the anterior choroidal fissure. The choroidal fissure is then opened and the hippocampus and fimbria are dissected free medially. Dissection then occurs under the hippocampus along the middle fossa floor and tentorium to the pial border of the tentorial incisura. The hippocampus is disconnected posteriorly at the level of the trigone and then rolled away from the parahippocampal gyrus and sent for pathologic examination. Lastly, the parahippocampal gyrus is removed in subpial fashion to the level of the posterior border of the brainstem. Cranial closure then occurs in standard fashion.

2.5.2. Laser interstitial thermal therapy

A Universal Compact Head Ring Assembly (UCHRA) system for the Cosman-Roberts-Wells (CRW) frame (Integra LifeSciences, Princeton, New Jersey, USA) is applied pre-operatively and the patient is transported for the computer tomography scanner. A computed tomography (CT) localizer frame is secured to the frame and a thin-slice CT obtained. This CT is co-registered with a preoperative MRI and stereotactic plan. Integra NeuroSight Arc software is used to create a trajectory along the long axis of the amygdala and hippocampus. The patient is then placed under general anesthesia and a Mayfield attachment is utilized to secure the UCHRA to the operating table. The CRW frame is tested against a Phantom Base, verifying its accuracy, and then secured to the UCHRA. An incision is planned based on the determined coordinates. A twist drill hole is made and an anchor bolt placed. The patient is then transported to the MRI suite where the laser is implanted, and proper position is confirmed. Real-time thermography and ablation with the Monteris NeuroBlate Optic FullFire probe (Monteris Medical, Winnipeg, Manitoba, Canada) is used to cover the desired structures. Once ablated, the laser and anchor bolt are removed, and the incision is closed with staples.

3. Results

3.1. Data characteristics

Patient demographics, costs, surgical variables, and hospitalization variables are outlined in Table 1 There were no statistically significant differences in patient demographics between groups ($p > 0.05$). The first and second columns show the properties of the data for all 59 patients undergoing ATL and all 40 patients undergoing LITT, respectively. The column labelled LITT ± ATL includes all LITT patients, but for the 14 LITT patients who also underwent a subsequent ATL, the values included in the sample are the combined totals across both surgeries. In this case the averages are computed per patient (not per surgery). The final column labelled LITT&ATL shows the data only for those 14 LITT patients who underwent a subsequent ATL, and the averages are again per patient. In our sample, 35 % of all LITT patients underwent a subsequent ATL procedure. Of note, none of the patients receiving ATL initially underwent a subsequent procedure.

3.2. Analysis of normal distributions

The results presented in Table 2 show that for all variables among

Table 1
Patient Demographics, Costs, Surgical Variables, and Hospitalization Variables.

		ATL	LITT	LITT ± ATL	LITT&ATL
Number of Surgeries		59	40	54	28
Number of Patients		59	40	40	14
Gender					
	Female	27	19	19	8
	Male	32	21	21	6
Age					
	Mean	39.7	38.1	38.4	37.1
	Standard deviation	13.3	12.2	11.8	10.8
Insurance Carrier					
	Medicaid	10	7	8	2
	Medicare	14	9	13	6
	Commercial	28	13	19	11
	Managed Care	6	7	9	6
	Other	1	1	1	0
Total Charges					
	Mean	$88,548	$82,176	$103,700	$145,084
	Standard deviation	$59,460	$24,755	$36,336	$13,211
	Maximum	$499,837	$144,295	$165,284	$165,284
	Minimum	$48,033	$33,397	$33,397	$127,563
	Median	$77,711	$79,956	$92,101	$140,120
Total Direct Costs					
	Mean	$14,256	$18,477	$21,442	$29,879
	Standard deviation	$13,814	$6,681	$8,579	$4,034
	Maximum	$113,107	$39,637	$39,637	$35,118
	Minimum	$5,402	$5,348	$5,348	$22,017
	Median	$12,558	$19,919	$20,782	$30,653
Peri-Op/Anesthesia Cost					
	Mean	$5,659	$4,257	$5,621	$8,434
	Standard deviation	$3,475	$2,385	$3,465	$3,317
Medical Supplies Direct Cost					
	Mean	$2,500	$12,135	$12,697	$15,905
	Standard deviation	$2,328	$6,212	$6,486	$4,922
Room, Board, & ICU Direct Cost					
	Mean	$3,573	$1,141	$1,768	$3,270
	Standard deviation	$6,396	$784	$1,314	$849
Length of Stay (LOS)					
	Mean (days)	3.37	1.19	1.86	3.45
	Standard deviation	4.01	.71	1.38	1.04
ICU Stay					
	Mean (days)	2.64	.95	1.51	2.82
	Standard deviation	3.90	.85	1.28	.98
LOS Index (LOS/GLOS)					
	Mean	.97	.45	.51	.60
	Standard deviation	.52	.16	.19	.21

Notes: ATL is single procedure data for all patients undergoing anterior temporal lobectomy; LITT is single procedure data for all patients undergoing laser interstitial thermal therapy; LITT ± ATL includes data for all 40 LITT patients, and for the 14 patients who had subsequent ATL, the values included are the total combined cost, surgical, and hospitalization values across both procedures; LITT&ATL includes data only for the 14 patients who had LITT and subsequent ATL, and the values included are the total combined cost, surgical, and hospitalization values across both procedures; GLOS (geometric length of stay) is the national mean length of stay for each diagnostic related group (DRG) as determined and published by CMS and represents the number of days the patient is expected to be hospitalized.

both ATL and LITT groups the Shapiro-Wilk test rejects that distributions of data are normal. Some of the variables when examining the combined data are normally distributed. However, because it is necessary to test the difference in means against the ATL sample, which is not normally distributed, a Mann–Whitney *U* test was employed.

Table 2
Shapiro-Wilk Normality Test Results.

Variable	Sample	W-stat	p-value	Normal?
Total Charges				
	ATL	.407	<.001*	no
	LITT	.913	.007*	no
	LITT ± ATL	.943	.059	yes
	LITT&ATL	.911	.250	yes
Total Direct Cost				
	ATL	.372	<.001*	no
	LITT	.840	<.001*	no
	LITT ± ATL	.962	.232	yes
	LITT&ATL	.944	.566	yes
Room, Board, & ICU Direct Cost				
	ATL	.276	<.001*	no
	LITT	.595	<.001*	no
	LITT ± ATL	.833	<.001*	no
	LITT&ATL	.887	.129	yes
Periop/Anesthesia Direct Cost				
	ATL	.914	<.001	no
	LITT	.927	.018	no
	LITT ± ATL	.963	.242	yes
	LITT&ATL	.910	.246	yes
Med Supplies Direct Cost				
	ATL	.507	<.001	no
	LITT	.866	<.001	no
	LITT ± ATL	.907	.005	no
	LITT&ATL	.570	<.001	no
Length of Stay (LOS)				
	ATL	.366	<.001*	no
	LITT	.305	<.001*	no
	LITT ± ATL	.717	<.001*	no
	LITT&ATL	.806	.011*	no
ICU days				
	ATL	.373	<.001*	no
	LITT	.579	<.001*	no
	LITT ± ATL	.825	<.001*	no
	LITT&ATL	.896	.165	yes
LOS Index (LOS/GLOS)				
	ATL	.868	<.001*	no
	LITT	.888	.002*	no
	LITT ± ATL	.948	.088	yes
	LITT&ATL	.947	.607	yes

Notes: ATL is single procedure data for all patients undergoing anterior temporal lobectomy; LITT is single procedure data for all patients undergoing laser interstitial thermal therapy; LITT ± ATL includes data for all 40 LITT patients, and for the 14 patients who had subsequent ATL, the values included are the total combined cost, surgical, and hospitalization values across both procedures; LITT&ATL includes data only for the 14 patients who had LITT and subsequent ATL, and the values included are the total combined cost, surgical, and hospitalization values across both procedures; * indicates statistical significance at the 5% level (or better) indicating the underlying data is not normally distributed.

3.3. Univariate analysis and difference in means tests for cost & hospitalization data

Table 3 presents the difference in the means of the variables across the samples of patients. In all cases the differences are computed relative to the ATL group. The first column shows the differences in the means relative to the initial LITT procedure. The second column, which centers on the contribution of our study, compares ATL versus the entire LITT sample once extra costs for those LITT patients who subsequently have an ATL (LITT ± ATL) are included. Of interest is how the comparisons of ATL with LITT in the first column (which are what has been done in previous studies) [3,15] are altered when one includes the cost of subsequent ATLs (the second column). The final column compares the values for ATL patients to only patients who have had both a LITT and ATL.

The average total charge for LITT-only patients is $6,372 lower than for ATL ($82,176 for LITT versus $88,548 for ATL), although this difference is not statistically significant ($p = .595$). Once the total charges data for the 14 subsequent ATLs are included, the average total charge

Table 3
Difference in Means Tests ATL versus LITT and ATL versus LITT with subsequent ATL.

	ATL vs. LITT	ATL vs. LITT ± ATL	ATL vs. LITT&ATL
Total Charges			
Mean difference	−$6,372	$15,152	$56,535
Mann-Whitney U z-score	.534	3.127	4.896
p-value (exact)	.595	.002*	<.001*
Total Direct Costs			
Mean difference	$4,221	$7,186	$15,624
Mann-Whitney U z-score	4.539	5.089	5.019
p-value (exact)	<.001*	<.001*	<.001*
Peri-Op/Anesthesia Cost			
Mean difference	−$1,401	−$38	$2,775
Mann-Whitney U z-score	1.588	.316	2.405
p-value (exact)	.112	.753	.015*
Medical Supplies Direct Cost			
Mean difference	$9,635	$10,197	$13,405
Mann-Whitney U z-score	6.286	6.459	4.486
p-value (exact)	<.001*	<.001*	<.001*
Room, Board, & ICU Direct Cost			
Mean difference	−$2,432	−$1,805	−$303
Mann-Whitney U z-score	7.225	4.120	2.076
p-value (exact)	<.001*	<.001*	.036*
Length of Stay			
Mean difference	−2.174	−1.452	.086
Mann-Whitney U z-score	7.267	3.863	2.153
p-value (exact)	<.001*	<.001*	.039*
ICU Stay			
Mean difference	−1.698	−1.131	.174
Mann-Whitney U z-score	5.538	2.880	2.030
p-value (exact)	<.001*	.005*	.050*
LOS Index (LOS/GLOS)			
Mean difference	−.514	−.455	−.370
Mann-Whitney U z-score	5.685	4.953	2.620
p-value (exact)	<.001*	<.001*	.008*

Notes: See notes to prior tables regarding ATL, LITT, LITT ± ATL, and LITT&ATL definitions and samples; * indicates statistical significance at the 5% level (or better) indicating the mean values are significantly different; all tests are two tailed; for each variable two difference in means tests are performed: the *t*-test, and the Mann–Whitney *U* test (also called the Mann–Whitney–Wilcoxon test or Wilcoxon rank-sum test) which is a nonparametric difference in means test robust to outliers and non-normal distributions.

rises to $103,700, which is $15,152 higher than the average total charges for ATL alone ($p < .001$). For the patients undergoing both LITT and ATL, the average total patient charges were $145,084, which is $56,535 greater than the average charges for an ATL ($p < .001$).

In contrast to total patient charges, the center's direct cost was actually $4,221 lower for ATL than LITT ($p < .001$). With the direct cost being higher for LITT than ATL for the procedures alone, once the subsequent ATL surgeries among LITT patients are included the direct cost difference grows even larger ($p < .001$).

By the direct cost measure, the average cost among all LITT patients after including the subsequent ATL procedures is $7,186 greater than the average cost of ATL surgery alone ($p < .001$). This is equal to about half the differential in the total charges. Among the subcategories of direct costs, the *peri*-op/anesthesia costs are not significantly different between the ATL and LITT groups ($p = .112$), even after calculating the average for all LITT patients including those who have subsequent ATL ($p = .753$).

The medical supplies direct cost is higher for LITT than ATL patients ($p < .001$) and grows even larger after including the subsequent ATL surgeries among LITT patients ($p < .001$). Room, board, and ICU direct costs are significantly lower for LITT patients than ATL patients, both with and without including the cost of the subsequent ATL procedures are included ($p < .001$).

Patient LOS in days is shorter for LITT than ATL patients, both with and without including the cost of the subsequent ATLs are included ($p < .001$), although as one would expect total combined LOS it is longer for those having both LITT and ATL than ATL alone ($p < .050$).

The LOS index (LOS/GLOS) compares a patient's actual LOS to their geometric length of stay (GLOS), which is the national mean LOS for each diagnostic related group as determined and published by the Centers for Medicare and Medicaid Services. In our sample, LITT patients, either with or without including the subsequent ATL, were released significantly earlier than expected relative to ATL patients ($p < .001$).

3.4. *Univariate analysis and difference in means tests for patient outcome data*

Table 4 shows the data for patient characteristics and outcomes among the groups, while Table 5 shows the difference in means tests for the proportion of seizure free patients among the surgical groups. The data indicated by 'weighted' is the number of seizure free months divided by total months observed, which more heavily weights patient data with longer follow up periods.

Total seizure freedom for ATL patients (81.8 %) was higher than LITT-only patients (27.0 %) ($p < .001$), and LITT patients with or without subsequent ATL (52.8 %) ($p = .003$). As expected, total seizure freedom among patients who had both a LITT and subsequent ATL was not significantly different from the ATL group ($p = .313$). The weighted measure of seizure freedom was also higher among ATL patients (82.3 %) compared to either LITT-only patients (33.7 %) ($p < .001$) or LITT patients with or without ATL (53.9 %) ($p < .001$). However, the difference is now also significant between patients who had both procedures versus those who had ATL only. The reason is that undergoing LITT delayed the subsequent date of seizure freedom for those patients who subsequently became seizure free after ATL. Those additional months spent having seizures after a failed LITT meant that the average LITT patient spent significantly less months being seizure free than the average patient for whom ATL was the first surgical treatment.

Lastly, from the data in Table 1 and 4 it is possible to derive the cost per seizure-free patient outcome, which is one of the variables from our economic cost model. Following our model, the cost per seizure-free patient outcome is derived by dividing the total charges by the seizure free rates. The results are $108,226 for ATL, $304,052 for LITT only, and $196,484 for LITT after accounting for the subsequent ATL surgeries. In our sample, ATL is significantly ($p < .001$), lower cost per seizure-free patient outcome than LITT, both excluding and including the follow up ATL procedures.

4. Discussion

Prior studies have compared the procedure costs of ATL and LITT and have generally concluded that LITT has a lower procedure cost. [3,6,15] These studies also note that one problem with the comparison is that some LITT patients undergo subsequent ATL due to the lower effectiveness of LITT and that "future emphasis will be placed on building a cost model that can account for the[se] costs" (p. e220). [3] In our study, we provide the first real-world evidence on the cost comparison between LITT and ATL incorporating the frequency and cost of ATL revision surgery among LITT patients.

Our economic cost model (Proposition 1) predicts that whenever the percentage of LITT patients who subsequently undergo ATL is greater than the procedure cost savings of LITT relative to ATL, the cost of these repeat surgeries will result in the average expected cost being higher for

Table 4
Patient Characteristics and Outcomes.

	ATL	LITT	LITT ± ATL	LITT&ATL
Number of Surgeries	55	37	50	26
Number of Patients	55	37	37	13
Gender				
Female	25	17	17	7
Male	30	20	20	6
Age				
Mean	39.9	38.4	38.6	38.2
Standard deviation	13.5	12.5	12.0	10.4
Age at Epilepsy Onset				
Mean	26.2	19.0	19.0	24.3
Standard deviation	14.7	12.6	12.6	13.2
Years with Epilepsy (at time of surgery)				
Mean	13.4	19.4	18.3	13.2
Standard deviation	12.1	15.9	15.7	13.7
Ictal scalp EEG				
Left	23	21	21	9
Right	15	13	13	3
Bilateral	3	3	3	1
Ictal intracranial EEG				
Left	8	11	11	4
Right	8	1	1	0
Bilateral	3	0	0	0
None	1	0	0	0
Initial MRI				
Normal	9	9	9	4
MTS	13	23	23	6
Hippocampal asymmetry	9	3	3	2
Other	11	2	2	1
Surgery and side				
Left ATL	33		9	9
Left LITT		24	24	9
Right ATL	22		4	4
Right LITT		13	13	4
Follow-up time (months)				
Mean	34.6	30.9	33.5	32.3
Standard deviation	21.6	22.5	22.1	20.5
Maximum	78	76	78	78
Minimum	6	4	4	4
Median	36	21	26	29.5
Engel Class				
I	45	10	19	9
II-IV	10	27	17	4
Seizure Freedom				
Total (percentage Engel Class 1)	81.8 %	27.0 %	52.8 %	69.2 %
Weighted (seizure free months / total months observed)	82.3 %	33.7 %	53.9 %	65.5 %

Notes: See notes to prior tables regarding ATL, LITT, LITT ± ATL, and LITT&ATL definitions and samples; only patients with outcome data and a minimum of 6 months follow-up period (unless early failure leading to repeat procedure) are included.

Table 5
Difference in Proportions Z-Test for Seizure Freedom Rates ATL versus LITT.

	ATL vs LITT	ATL vs LITT ± ATL	ATL vs LITT&ATL
Total (percentage Engel Class 1) Data:			
Seizure Freedom ATL Group	81.8 %	81.8 %	81.8 %
Seizure Freedom Comparison Group	27.0 %	52.8 %	69.2 %
Mean difference	−54.8 %	−29.0 %	−12.6 %
z-stat	5.255	2.965	1.009
p-value	<.001*	.003*	.313
Significantly Different?	yes	yes	no
Weighted (seizure free months / total months observed) Data:			
Seizure Freedom ATL Group	82.3 %	82.3 %	82.3 %
Seizure Freedom Comparison Group	33.7 %	53.9 %	65.5 %
Mean difference	−48.7 %	−28.4 %	−16.9 %
z-stat	27.115	17.522	8.379
p-value	<.001*	<.001*	<.001*
Significantly Different?	yes	yes	yes

Notes: See notes to prior tables regarding ATL, LITT, LITT ± ATL, and LITT&ATL definitions and samples; sample sizes for Total (percentage Engel Class 1) are 55 for ATL, 37 for LITT, 36 for LITT ± ATL, 13 for LITT&ATL; sample sizes for Weighted (seizure free months / total months observed) are 1902 for ATL, 1144 for LITT, 1357 for LITT ± ATL, and 530 for LITT&ATL; * indicates statistical significance at the 5% level (or better) indicating the mean proportions are significantly different; all tests are two tailed, two sample, difference in proportions Z tests.

patients whose first course of treatment is LITT relative to those whose first course of treatment is ATL. This is what we find to be the case in our sample.

Among the 40 LITT patients studied here, 35 % underwent subsequent ATL. Based upon patient total charges, the LITT procedure was only 7.2 % less costly than ATL ($82,176 for LITT versus $88,548 for ATL). The result is that after accounting for the subsequent surgeries, the average patient charge among the LITT sample ($103,700) was significantly higher than for the ATL sample ($88,548). The effect of accounting for the repeat interventions eliminates the cost advantage of LITT among the patents studied in our sample. Using hospital direct costs rather than patient charges produces an even larger cost advantage for ATL, as in our sample the direct cost was lower for ATL than LITT without any adjustment for follow up surgeries.

We recognize that the rates of subsequent ATL surgeries after LITT will vary by center, as will the differential in costs. Our economic model can be applied to any center specific numbers, or to numbers from other published studies. For example, other studies have computed the average procedure costs of LITT and ATL that can be used for comparison. Brandel, et. al. [15] found that the cost advantage of the LITT procedure is 20.8 % using total charges or 18.2 % using hospital costs, while Hines et al. [3] found a difference of 22.2 % excluding outliers or 9.1 % using all patients. These cost differentials are larger than within our sample (7.2 %), but are still not large enough to offset the additional cost of 35 % of LITT patients subsequently undergoing ATL.

The rate of post-LITT ATL procedures in our study (35 %) may be higher than other centers, although there are very few other studies that give statistics on this outcome. Youngerman et al. [10] found that out of 268 LITT patients, 36 underwent subsequent surgeries (13.4 %), 22 of which underwent subsequent ATL (8.2 %). This lower rate of subsequent ATL procedures lowers the cost differential necessary to meet the condition in our economic cost Proposition 1. However, even Youngerman, et. al.'s [10] lower 8.2 % rate would still exceed the cost savings in the sample from our center (7.2 %). These numbers could represent an underestimate of the true number of patients who undergo subsequent procedures, as it is possible that patients lost to follow up could be seen at other centers. Although this requires speculation, any subsequent procedures would not be included in the data, which represents an inherent flaw of the retrospective design of all current studies done examining costs and outcome. Additionally, a lower rate of ATL post-LITT may provide fewer seizure-free patients from follow-up ATL procedures, as the literature clearly finds that the effectiveness of LITT is lower than ATL. [7,16] Our second economic model proposition takes these rates of effectiveness into account by comparing the cost per seizure-free patient outcome. In our sample, the cost per seizure-free patient outcome is $108,226 for ATL, $304,052 for LITT only, and $196,484 for LITT after accounting for the subsequent ATL surgeries. Based upon that data, ATL as a first treatment is the more economically cost-effective means of achieving each seizure free patient.

The second proposition was restated in a way that makes it more applicable to using other estimates from the literature. This corollary compares the ratio of the effectiveness of LITT versus ATL to the ratio of the cost of the procedures. In our sample, the total seizure-free rate from LITT was 27.0 % while it was 81.8 % from ATL, yielding a ratio of 33.0 %. That number is 40.9 % if one uses the weighted months of seizure freedom observed among patients. For LITT to be more economically cost-effective, the ratio of the procedure costs should be lower than these numbers. In our sample it is not, and the most favorable (i.e., lowest) cost ratio for LITT is 92.8 % using total charges.

Using the cost numbers from the other studies, the cost ratios are 79.2 % (total charges) or 81.8 % (cost) from Brandel et al. [15] and 77.8 % excluding two outliers or 90.9 % including all patients in Hines et al. [3] In no case do these favor LITT as they remain much higher than the effectiveness ratios of 33.0 % or 40.9 % in our sample.

However, the effectiveness of LITT in achieving seizure freedom in our study is low compared to other published data. [7,16] Therefore, it is worth considering how the ratio comparison holds up to other studies that measure the seizure free rates from ATL and LITT. A *meta*-analysis of 32 published studies by Marathe et al. [7] found that the seizure free rate per person year associated with LITT is 59 %, compared to 70 % for ATL. This makes LITT about 84.3 % as effective as ATL. Even if this higher effectiveness ratio for LITT had held in our sample, it would not be enough to offset our cost ratio of 90.9 %, although it would offset the one of the two cost ratios (the one that excludes outliers) from Hines et al. [3] and be roughly equal to the cost ratios from Brandel et al. [15] Furthermore, while our seizure-free rate of 27 % for LiTT appears low in comparison to other data, it is important to note that our median follow-up time was 30.9 months. Based on logistic regression from a *meta*-analysis by Chen et al [17], seizure-freedom from LiTT in patients with MTS was about 40 % at this time and was even lower across their study for patients with normal MRIs or generalized seizures. In contrast, Pereira Dalio et al [18] found maintained seizure-freedom following ATL in 65 % of patients with TLE with hippocampal sclerosis even after 20 years, suggesting that the duration of follow-up may be a significant contributing factor. In considering the low rates of seizure-freedom for LITT in our study, it is fair to consider the novelty of LITT compared to traditional ATL. It is understandable that as more experience with targeted and optimized ablations is obtained, there may be a rise in seizure-freedom rates. This is a possible contributing factor to our lower seizure-freedom rates as our data consists of patients treated from 2015 to 2022.

Length of ICU stay is a factor that can increase hospitalization cost of all groups. In our study, the mean ICU stay for ATL and LITT was 2.64 and 0.95 days, respectively. Data on typical practice and protocols in this regard seems to be rather limited. Hines et al. [3] has a total LOS of 1.1 days for LITT, while our LITT population was 1.19. ICU costs in Hines et al. [3] was counted in post procedural costs, which were $16,900 for ATL and $7,200 for LITT. The ratio of these costs is the approximate ratio in costs compared to the number of days spent in the ICU for each procedure in our study.

We conclude that for a wide range of estimates from the prior literature, the economic cost models presented here suggest that not only is LITT less effective in achieving seizure freedom, but that once the cost of subsequent follow-up ATL procedures is included, the expected patient cost is significantly higher for LITT as the first course of treatment based on many of the aforementioned permutations.

Despite its lower effectiveness and higher expected cost, LITT may remain a viable treatment option for many patients for other reasons. Since it is less invasive, more patients may be willing to undergo the procedure, and there is a lower risk of complications [7,19] if LITT does result in seizure freedom (although this advantage is eliminated if the patient undergoes both LITT and subsequent ATL). For patients with dominant mTLE, LITT may also be more desirable as attempt to preserve language or verbal memory since there is some evidence that LITT may be associated with less cognitive decline in these domains. [8,20] On the other hand, for patients with structural lesions requiring pathologic diagnosis, ATL may be necessary for making this determination regardless of costs, and the outcomes for these patients may be satisfactory even with less extensive resection. [21].

An important consideration would be a projected financial burden on the patient based on re-admission rates and ER visits for epilepsy related complications. Although this is outside the scope of this study, previous literature would support that a significant financial impact for these patients is uncontrolled epilepsy. [23] Therefore, it would be anticipated that the financial burden would be reduced in a population undergoing a procedure with higher rates of seizure-freedom.

4.1. *Strengths and limitations*

Our study has several strengths over previous studies. Most notably, we account for repeat interventions in our cost comparison, which no study to this point has done. We also utilize real-world data, which provides a more accurate account of clinical factors in comparison to potentially larger administrative data sets, which are prone to coding errors. However, the use of single-center data could mean our results are not predictive of the costs or outcomes at other surgical centers leading to the potential for reduced external validity. This limitation is somewhat nullified by our employment of a model for use by individual institutions to calculate differences in costs, outcomes, and other hospital measures. Our use of administrative data could suffer from coding errors or other limitations as the data is intended for billing purposes. While seizure-freedom is a worthy metric for assessing cost utility, our study does not examine costs in comparison to other neuropsychological, morbidity, or quality of life data. Additionally, our study does not account for complications necessitating re-hospitalization or subsequent ER visits for either LITT or ATL. Further studies to examine these factors could further elucidate the total costs and benefits associated with different surgical approaches.

5. Conclusion

In this study, we provide the first cost comparison between LITT and ATL incorporating the frequency and cost of subsequent surgeries. We also use an economic model to produce a surgical decision-making framework for understanding what levels of cost differentials, relative surgical effectiveness, and rates of revision surgery result in ATL having cost advantages to LITT, and vice versa. In our sample, after accounting for the subsequent post-LITT ATL surgeries, the average patient charge for LITT was significantly higher than for ATL, as was the cost per seizure-free patient outcome.

The cost advantage of LITT previously assumed in the literature based on lower procedure cost does not necessarily hold true once the subsequent ATL surgeries are accounted for among LITT patients. In some scenarios including at our center, ATL as the first course of treatment yields both the least expected patient cost and is more effective in generating seizure free outcomes, although it may be a less desirable course of action for other reasons. Perhaps more importantly, the cost-benefit may vary across other institutions, which can utilize our model to make that determination based on local data. Further multicenter studies implementing our model could be useful at gauging larger overall trends.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1A. Proof of Proposition 1

Let N_L be the number of patients undergoing LITT, N_{L+A} the number of LITT patients undergoing a subsequent ATL, C_A the average cost of an ATL procedure, and C_L the average cost of a LITT procedure.

The total combined cost for all LITT patients (TC_L) including those who undergo the subsequent ATL is equal to the cost of the LITT procedures plus the cost of any subsequent ATL procedures among the group. With N_L being the number of patients undergoing LITT, and C_L being the average cost of a LITT procedure, the total cost of all the LITT procedures is $N_L \times C_L$. With N_{L+A} being the number of patients undergoing subsequent ATL after LITT, and C_A being the average cost of an ATL procedure, the total cost of all the subsequent ATL procedures is $N_{L+A} \times C_A$. Thus, the combined total cost for all LITT patients of their initial LITT procedures plus the additional costs of the subsequent ATL procedures among the subset of patients who undergo them (TC_L) is thus $N_L \times C_L + N_{L+A} \times C_A$. The average cost per patient can be obtained by dividing this total cost by the number of patients, which is TC_L / N_L, or through substitution: $(N_L \times C_L + N_{L+A} \times C_A) / N_L$, which can be reduced to: $C_L + [(N_{L+A} / N_L) \times C_A]$.

Because the average cost among patients having only an ATL is C_A, the average cost per patient among LITT patients (including subsequent ATL procedures) will be ***higher*** than the average cost among ATL (only) patients if $C_L + [(N_{L+A} / N_L) \times C_A] > C_A$. Through manipulation this is equivalent to the condition presented in Proposition 1 that N_{L+A}?/ $N_L > [(C_A – C_L) / C_A]$. Verbally this condition states that the average cost per patient among LITT patients (including subsequent ATL procedures) will be ***higher*** than the average cost among ATL (only) patients if the proportion of LITT patients undergoing ATL (which is given by N_{L+A} / N_L) exceeds the percentage by which LITT procedure costs are lower than ATL procedure costs (which is given by $(C_A – C_L) / C_A$).

Appendix 1B. Proof of Proposition 2

Let SF_A be the seizure-free rate associated with ATL and let SF_L be the seizure free rate associated with LITT. Let N be the total number of patients. As before let C_A be the average cost of an ATL procedure, and C_L the average cost of a LITT procedure. If all patients undergo ATL, the expected number of seizure-free patients would be equal to $N \times SF_A$, and the total cost would be equal to $N \times C_A$.

If all patients undergo LITT, the expected number of seizure-free patients from the LITT alone would be equal to $N \times SF_L$. If N_{L+A} of the N patients undergoing LITT then undergo a subsequent ATL, among that group having the subsequent ATL we would expect an additional $N_{L+A} \times SF_A$ to become seizure free from the ATL procedure. Thus, the total number of the LITT patients expected to become seizure free in the end is equal to $N \times SF_L + N_{L+A} \times SF_A$.

The two courses of surgical action will generate the same number of seizure-free patients when: $N \times SF_A = N \times SF_L + N_{L+A} \times SF_A$, which will occur only if a specific number of LITT patients undergo subsequent ATL. This can be found through manipulation, which produces $N_{L+A} = N \times [(SF_A – SF_L) / SF_A]$.

The total cost for sending all patients through LITT first, with N_{L+A} patients having a subsequent ATL is: $TC_L = N \times C_L + N_{L+A} \times C_A$. Filling in for N_{L+A} based on our above derivation of how many are required to achieve the same outcome effectiveness yields: $TC_L = N \times C_L + [N \times (SF_A – SF_L) / SF_A] \times C_A$.

Because we have adjusted the proportions to achieve the same number of seizure-free patients, we are interested in which produces that number at a lower total cost. ATL will be less expensive (and the LITT route more expensive) if: $N \times C_A < N \times C_L + [N \times (SF_A – SF_L) / SF_A] \times C_A$. Continuing to simplify yields $C_A < C_L + [(SF_A – SF_L) / SF_A] \times C_A$, or $C_A \times (SF_L / SF_A) < C_L$, which can be written as:

$$\frac{C_A}{SF_A} < \frac{C_L}{SF_L} \text{or, equivalently, } \frac{SF_L}{SF_A} < \frac{C_L}{C_A}$$

It is worth explicitly noting that the economic principle of duality [22] implies that the same decision-making condition that minimizes cost for a given number of seizure-free patients in a surgical population also equivalently produces the highest number of seizure-free patients for a given level of total system-wide surgical cost. We omit the derivation, but it produces identical results.

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